Lecture 10: Duality

MATH 110-3

Franny Dean

July 6, 2023

Notation

Let V be a $\mathbb{F}$-vector space.

Linear Functionals

Def'n:

A **linear functional** on V is a linear map from V to $\mathbb{F}$, i.e. an element of $\mathcal{L}(V, \mathbb{F})$.

Linear Functionals

Def'n:

A **linear functional** on V is a linear map from V to $\mathbb{F}$, i.e. an element of $\mathcal{L}(V, \mathbb{F})$.

Examples:

Linear Functionals

Def'n:

A **linear functional** on V is a linear map from V to $\mathbb{F}$, i.e. an element of $\mathcal{L}(V, \mathbb{F})$.

Examples:

- $\phi : \mathbb{R}^3 \to \mathbb{R}$ where $\phi(x, y, z) = 6x + y - z$

Linear Functionals

Def'n:

A **linear functional** on V is a linear map from V to $\mathbb{F}$, i.e. an element of $\mathcal{L}(V, \mathbb{F})$.

Examples:

- $\phi : \mathbb{R}^3 \to \mathbb{R}$ where $\phi(x, y, z) = 6x + y - z$
- $\phi : \mathbb{F}^n \to \mathbb{F}$ where $\phi(x_1, \ldots, x_n) = c_1x_1 + \ldots + c_nx_n$

Linear Functionals

Def'n:

A **linear functional** on V is a linear map from V to $\mathbb{F}$, i.e. an element of $\mathcal{L}(V, \mathbb{F})$.

Examples:

- $\phi : \mathbb{R}^3 \to \mathbb{R}$ where $\phi(x, y, z) = 6x + y - z$
- $\phi : \mathbb{F}^n \to \mathbb{F}$ where $\phi(x_1, \ldots, x_n) = c_1x_1 + \ldots + c_nx_n$
- $\phi : \mathcal{P}(\mathbb{R}) \to \mathbb{R}$ where $\phi(p) = \int_0^1 p dx$

Dual Space

Def'n:

We call $\mathcal{L}(V, \mathbb{F})$ the **dual space** of V and denote it V'.

Dual Space

Def'n:

We call $\mathcal{L}(V, \mathbb{F})$ the **dual space** of V and denote it V'.

Sometimes V' is denoted V^* elsewhere.

Dual Space

Def'n:

We call $\mathcal{L}(V, \mathbb{F})$ the **dual space** of V and denote it V'.

Sometimes V' is denoted V^* elsewhere.

Prop'n:

For V finite-dimensional, V' is also finite-dimensional and $\dim V' = \dim V$.

Dual Space

Prop'n:

For V finite-dimensional, V' is also finite-dimensional and $\dim V' = \dim V$.

Dual Space

Prop'n:

For V finite-dimensional, V' is also finite-dimensional and $\dim V' = \dim V$.

Proof.

Dual Space

Prop'n:

For V finite-dimensional, V' is also finite-dimensional and $\dim V' = \dim V$.

Proof. Recall that $\mathcal{L}(V, W) \cong \mathbb{F}^{m,n}$.

Dual Space

Prop'n:

For V finite-dimensional, V' is also finite-dimensional and $\dim V' = \dim V$.

Proof. Recall that $\mathcal{L}(V, W) \cong \mathbb{F}^{m,n}$.

Then $\dim \mathcal{L}(V, W) = \dim \mathbb{F}^{n,m} = \dim V \cdot \dim W$.

Dual Space

Prop'n:

For V finite-dimensional, V' is also finite-dimensional and $\dim V' = \dim V$.

Proof. Recall that $\mathcal{L}(V, W) \cong \mathbb{F}^{m,n}$.

Then $\dim \mathcal{L}(V, W) = \dim \mathbb{F}^{n,m} = \dim V \cdot \dim W$.

$\dim \mathcal{L}(V, \mathbb{F}) = \dim V \cdot \dim \mathbb{F} = \dim V$.

Dual Basis

Def'n:

If $v_1, \ldots, v_n$ is a basis of V, then the dual basis of $v_1, \ldots, v_n$ is the list of $\phi_1, \ldots, \phi_n$ of V' such that

$$\phi_j(v_k) = \begin{cases} 1 & k = j, \\ 0 & k \neq j \end{cases}$$

Dual Basis

Def'n:

If $v_1, \ldots, v_n$ is a basis of V, then the dual basis of $v_1, \ldots, v_n$ is the list of $\phi_1, \ldots, \phi_n$ of V' such that

$$\phi_j(v_k) = \begin{cases} 1 & k = j, \\ 0 & k \neq j \end{cases}$$

Examples:

- Dual basis of $e_1, \ldots, e_n$?

Dual Basis

Def'n:

If $v_1, \ldots, v_n$ is a basis of V, then the dual basis of $v_1, \ldots, v_n$ is the list of $\phi_1, \ldots, \phi_n$ of V' such that

$$\phi_j(v_k) = \begin{cases} 1 & k = j, \\ 0 & k \neq j \end{cases}$$

Examples:

- Dual basis of $e_1, \ldots, e_n$?
- $V = \mathbb{R}^2$, $B = \{(2,1),(3,1)$. Find the dual basis.

Dual Basis

Prop'n:

For V finite-dimensional, the dual basis of V is a basis of V'.

Dual Basis

Prop'n:

For V finite-dimensional, the dual basis of V is a basis of V'.

Proof.

Dual Basis

Prop'n:

For V finite-dimensional, the dual basis of V is a basis of V'.

Proof. S'pose $v_1, \ldots, v_n$ is a basis of V and $\phi_1, \ldots, \phi_n$ denotes the dual basis.

Dual Basis

Prop'n:

For V finite-dimensional, the dual basis of V is a basis of V'.

Proof. S'pose $v_1, \ldots, v_n$ is a basis of V and $\phi_1, \ldots, \phi_n$ denotes the dual basis. It suffices to show that $\phi_1, \ldots, \phi_n$ is linearly independent.

Let

$$a_1\phi_1 + \ldots + a_n\phi_n = 0.$$

Dual Basis

Prop'n:

For V finite-dimensional, the dual basis of V is a basis of V'.

Proof. S'pose $v_1, \ldots, v_n$ is a basis of V and $\phi_1, \ldots, \phi_n$ denotes the dual basis. It suffices to show that $\phi_1, \ldots, \phi_n$ is linearly independent.

Let

$$a_1\phi_1 + \ldots + a_n\phi_n = 0.$$

Notice $(a_1\phi_1 + \ldots + a_n\phi_n)(v_j) = a_j$ for each $j \in [n]$.

Dual Basis

Prop'n:

For V finite-dimensional, the dual basis of V is a basis of V'.

Proof. S'pose $v_1, \ldots, v_n$ is a basis of V and $\phi_1, \ldots, \phi_n$ denotes the dual basis. It suffices to show that $\phi_1, \ldots, \phi_n$ is linearly independent.

Let

$$a_1\phi_1 + \ldots + a_n\phi_n = 0.$$

Notice $(a_1\phi_1 + \ldots + a_n\phi_n)(v_j) = a_j$ for each $j \in [n]$. Thus, $a_j = 0$ for each j. $\square$

Dual Map

Def'n:

If $T \in \mathcal{L}(V, W)$, the **dual map** is the linear map $T' \in \mathcal{L}(W', V')$ defined $T'(\phi) = \phi \circ T$ for $\phi \in W'$.

Dual Map

Def'n:

If $T \in \mathcal{L}(V, W)$, the **dual map** is the linear map $T' \in \mathcal{L}(W', V')$ defined $T'(\phi) = \phi \circ T$ for $\phi \in W'$.

Examples:

Dual Map

Def'n:

If $T \in \mathcal{L}(V, W)$, the **dual map** is the linear map $T' \in \mathcal{L}(W', V')$ defined $T'(\phi) = \phi \circ T$ for $\phi \in W'$.

Examples:

Let $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be the differentiation map $p \to p'$.

Dual Map

Def'n:

If $T \in \mathcal{L}(V, W)$, the **dual map** is the linear map $T' \in \mathcal{L}(W', V')$ defined $T'(\phi) = \phi \circ T$ for $\phi \in W'$.

Examples:

Let $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be the differentiation map $p \to p'$.

D' is the dual map from $\mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R}) \to \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$.

Dual Map

Def'n:

If $T \in \mathcal{L}(V, W)$, the **dual map** is the linear map $T' \in \mathcal{L}(W', V')$ defined $T'(\phi) = \phi \circ T$ for $\phi \in W'$.

Examples:

Let $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be the differentiation map $p \to p'$.

D' is the dual map from $\mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R}) \to \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$.

For example, if $\phi \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined to be $\phi(p) = p(3)$.

Dual Map

Def'n:

If $T \in \mathcal{L}(V, W)$, the **dual map** is the linear map $T' \in \mathcal{L}(W', V')$ defined $T'(\phi) = \phi \circ T$ for $\phi \in W'$.

Examples:

Let $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be the differentiation map $p \to p'$.

D' is the dual map from $\mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R}) \to \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$.

For example, if $\phi \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined to be $\phi(p) = p(3)$.

Then $D'(\phi)(p) = \phi(D(p)) = \phi(p') = p'(3)$.

Dual Map

Def'n:

If $T \in \mathcal{L}(V, W)$, the **dual map** is the linear map $T' \in \mathcal{L}(W', V')$ defined $T'(\phi) = \phi \circ T$ for $\phi \in W'$.

Examples:

Let $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be the differentiation map $p \to p'$.

D' is the dual map from $\mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R}) \to \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$.

For example, if $\phi \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined to be $\phi(p) = p(3)$.

Then $D'(\phi)(p) = \phi(D(p)) = \phi(p') = p'(3)$.

If $\phi \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined to be $\phi(p) = \int_0^1 p$.

Dual Map

Def'n:

If $T \in \mathcal{L}(V, W)$, the **dual map** is the linear map $T' \in \mathcal{L}(W', V')$ defined $T'(\phi) = \phi \circ T$ for $\phi \in W'$.

Examples:

Let $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be the differentiation map $p \to p'$.

D' is the dual map from $\mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R}) \to \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$.

For example, if $\phi \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined to be $\phi(p) = p(3)$.

Then $D'(\phi)(p) = \phi(D(p)) = \phi(p') = p'(3)$.

If $\phi \in \mathcal{L}(\mathcal{P}(\mathbb{R}), \mathbb{R})$ defined to be $\phi(p) = \int_0^1 p$.

Then $D'(\phi)(p) = \phi(D(p)) = \phi(p') = \int_0^1 p' = p(1) - p(0)$.

Dual Map

Facts about Dual Map:

- $(S+T)' = S' + T'$ for all $S, T \in \mathcal{L}(V, W)$
- $(\lambda T)' = \lambda T'$ for all $\lambda \in \mathbb{F}$ and $T \in \mathcal{L}(V, W)$
- $(ST)' = T'S'$ for all $T \in \mathcal{L}(U, v)$ and all $S \in \mathcal{L}(V, W)$

Examples and Exercises

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T(x, y, z) = (10x + 8y + 2z, x + y - z)$.

Examples and Exercises

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T(x, y, z) = (10x + 8y + 2z, x + y - z)$.

Suppose ϕ_1, ϕ_2 denotes the dual basis of the standard basis of $\mathbb{R}^2$.

What are the linear functionals $T'(\phi_1)$ and $T'(\phi_2)$?

Examples and Exercises

Define $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ by $(Tp)(x) = x^2 p(x)$ for $x \in \mathbb{R}$.

Examples and Exercises

Define $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ by $(Tp)(x) = x^2 p(x)$ for $x \in \mathbb{R}$.

Suppose $\phi \in \mathcal{P}(\mathbb{R})'$ defined by $\phi(p) = p''(7)$.

Examples and Exercises

Define $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ by $(Tp)(x) = x^2 p(x)$ for $x \in \mathbb{R}$.

Suppose $\phi \in \mathcal{P}(\mathbb{R})'$ defined by $\phi(p) = p''(7)$.

Describe $T'(\phi)$ on $\mathcal{P}(\mathbb{R})$.

Examples and Exercises

Suppose V is finite-dimensional and U is a subspace of V such that $U \neq V$. Prove that there exists $\phi \in V'$ such that $\phi(u) = 0$ for every $u \in U$ but $\phi \neq 0$.

Examples and Exercises

Suppose V is finite dimensional and $v \in V$ with $v \neq 0$. Prove that there exists $\phi \in V'$ such that $\phi(v) = 1$.

References

[Axl14] Sheldon Axler.
Linear Algebra Done Right.
Undergraduate Texts in Mathematics. Springer Cham, 2014.